

# Using HPGL as a drawing interchange media

(Don't! Unfortunately many DOS/Windows insist on doing so)

Written by [Paul Bourke](#)

September 1992

This document is also [available in Russian](#), thanks to Sergey Paramonov.

(You will need the Russian font set!)

## Introduction

HPGL is the HP graphics language designed for specifying 2D graphical information for the HP range of plotters. It was not intended as an interchange format but some software suppliers use it as such. At least it is efficient and easy to write and read. This short note describes those aspects of the format which may be used to describe geometry. The language itself has a large number of instructions, most of the ones not touched on here relate directly to plotter devices, for a full description of the language the HPGL description is normally provided with an HPGL compatible plotter.

## Basics

HPGL commands are basically two upper-case character instructions followed by any arguments, and finally ending in the separator character which is a semicolon by default. The arguments are normally separated by commas. String arguments are normally terminated by carriage return.

In what follows, all the standard HPGL commands are listed, those which are likely to be useful for data interchange will be discussed at the end of this document. Optional arguments for any command are shown in brackets.

## Commands

|    | Instruction                | Arguments                        |
|----|----------------------------|----------------------------------|
| AA | Arc Absolute               | x,y,angle(,tolerance)            |
| AF | Advance page               |                                  |
| AH | Advance page               |                                  |
| AP | Automatic Pen Operations   | n                                |
| AR | Arc Relative               | x,y,angle(,tolerance)            |
| AS | Acceleration Select        | acceleration(,pennumber)         |
| BF | Buffer Plot                |                                  |
| BL | Buffer label               | string                           |
| CA | Select Alternative Charset | set                              |
| CC | Character chord angle      | angle                            |
| CI | Circle                     | radius(,tolerance)               |
| CM | Character Selection Mode   | switchmode(,fallbackmode)        |
| CP | Character Plot             | spaces,lines                     |
| CS | Select Standard Charset    | set                              |
| CT | Chord Tolerance            | n                                |
| CV | Curved line generator      | n(,inputdelay)                   |
| CD | Digitize Clear             |                                  |
| DF | Default                    |                                  |
| DI | Absolute Direction         | run,rise                         |
| DL | Define Download character  | charnumber(,pencontrol),x,y(...) |
| DP | Digitize Point             |                                  |
| DR | Relative Direction         | run.rise                         |

|    |                            |                                                        |
|----|----------------------------|--------------------------------------------------------|
| DS | Designate Charset          | slot,set                                               |
| DT | Define Label Terminator    | terminator                                             |
| EA | Edge Rectangle Absolute    | x,y                                                    |
| EP | Edge Polygon               |                                                        |
| ER | Edge Rectangle Relative    | x,y                                                    |
| ES | Extra Space                | spaces(,lines)                                         |
| EW | Edge Wedge                 | radius,startangle,sweepangle(,tolerance)               |
| FP | Fill Polygon               |                                                        |
| FS | Force Select               | force(.pennumber)                                      |
| FT | Fill Type                  | type(,spacing(,angle))                                 |
| GC | Group Count                | countnumber                                            |
| GM | Graphics Memory            | (polygonbuff)(,downloadbuff)(,replotbuff)(,vectorbuff) |
| GP | Group Pen                  | (groupnumber(,pennumber(,numberofpens,(length))))      |
| IM | Input Mask                 | emaskvalue(,smaskvalue(,pmaskvalue))                   |
| IN | Initialize                 |                                                        |
| IP | Input p1 and p2            | x,y(,x,y)                                              |
| IV | Invoke Character Slot      | slot,(left)                                            |
| IW | Input Window               | x,y,x,y                                                |
| KY | Define Key                 | key(,function)                                         |
| LB | Label                      | string                                                 |
| LO | Label Origin               | positionnumber                                         |
| LT | Line Type                  | patternnumber(,patternlength)                          |
| NR | Not Ready                  |                                                        |
| PA | Plot Absolute              | x,y(,...)                                              |
| PB | Print Buffered Label       |                                                        |
| PD | Pen Down                   | x,y(,...)                                              |
| PG | Page Feed                  | n                                                      |
| PM | Polygon Mode               | n                                                      |
| PR | Plot Relative              | x,y(,...)                                              |
| PT | Pen Thickness              | thickness                                              |
| PU | Pen Up                     | x,y(,...)                                              |
| RA | Fill Rectangle Absolute    | x,y                                                    |
| RO | Rotate Coordinate System   | n                                                      |
| RP | Replot                     | n                                                      |
| RR | Fill Rectangle Relative    | x,y                                                    |
| SA | Select Alternative Charset |                                                        |
| SC | Scale                      | x,y,x,y(,type(,left,bottom))                           |
| SG | Select Pen Group           | groupnumber                                            |
| SI | Absolute Character Size    | width,height                                           |
| SL | Character Slant            | tantheta                                               |
| SM | Symbol Mode                | character                                              |
| SP | Select Pen                 | pennumber                                              |
| SR | Relative Character Size    | width,height                                           |
| SS | Select Standard Charset    |                                                        |
| TL | Tick Length                | tp(,tn)                                                |
| UC | User Defined Character     | (pencontrol,)x,y                                       |
| UF | User Defined Fill          | gap1(,gap2,...gap20)                                   |
| VS | Velocity Select            | speed(,pennumber)                                      |
| WD | Write to Display           | string                                                 |
| WG | Fill Wedge                 | radius,startangle,sweepangle(,tolerance)               |
| XT | X Tick                     |                                                        |
| YT | Y Tick                     |                                                        |

## Output Instructions

The following instructions don't have any arguments but rather generate output from the plotting device. They are of no use for interchanging geometric data and therefore the form of the output is not discussed. See an HPGL reference book.

|    |                        |
|----|------------------------|
| OA | Output Actual Position |
| OC | Output Position        |
| OD | Output Digitised Point |
| OE | Output Error           |

OF Output Factors  
OG Output Group Count  
OH Output HardClip Limits  
OI Output Identification  
OK Output Key  
OL Output Label Length  
OO Output Options  
OP Output p1 and p2  
OS Output Status  
OT Output Carousel Type  
OW Output Window

## Escape sequences

The following escape sequences are generally only applicable when connected through a serial interface to a physical plotter, they have nothing to do with data interchange and so their arguments will not be listed nor will the form of the output most of them create .

esc@ Set Plotter Configuration  
escA Output Identification  
escB Output Buffer Space  
escE Output Extended error  
escH Set Handshake Mode 1  
escI Set Handshake Mode 2  
escJ Abort Device Control  
escK Abort Graphics  
escL Output Buffer Size When Empty  
escM Set Output Mode  
escN Set Extended Output and Handshake Mode  
escO Output Extended Status  
escP Set Handshake Mode  
escQ Set Monitor Mode  
escR Reset  
escS Output Configurable Memory  
escT Allocate Configurable Memory  
escU End Flush Mode  
escY Plotter On  
esc( Plotter On  
escZ Plotter Off  
esc) Plotter Off

## Useful Commands with examples

Since plotters are very basic line drawing hardware devices it is not surprising that the most commonly used commands are those that draw lines from one point to another.

This operation is controlled by 4 commands,

**PA** - plot absolute

**PR** - plot relative

**PU** - pen up

**PD** - pen down

Pen up and down dictate when a line is actually being created, the plot commands do the movement between points.

For example to draw a line from (1,0) to (2,0) might be specified as follows

PU;PA1,0;PD;Pr1,0;PU;

There are two basic attributes that can be assigned to a line, as expected they are actually applied to the pens of the plotting device. They are:

**SP** - pen number

**PT** - pen thickness

The pen number normally acts as an indexed colour system, most commonly there are only 8 pens (colour) available.

The other geometry that can be specified is

**AR** - arc relative

**CI** - circle

**EA** - edge rectangle absolute

**ER** - edge rectangle relative

**EP** - edge polygon

**EW** - edge wedge

**LB** - label

There are a large number of settings for controlling text, these and further details of the above instructions will be left to the reader to experiment with or read in the full HPGL manuals.